

Experiment :— study of LAN Topology, Network devices and Address configuration.

Objective :—

- 1.) To study the LAN topology used in college campus.
- 2.) To identify networking device & understand their function.
- 3.) To analyze wired & wireless communication method used in the network.
- 4.) To measure speed of Network for wired & wireless connection.
- 5.) To obtain MAC, IP Address & subnet mask.

→ Problem Statement - I

- Draw & identify LAN Topology of college campus & clearly label all components such as router, switches & etc.

Soln :— The college campus network observed this study consists of :

- * Layer-3 core switches in the server room.
- * Managed by POE + switches.
- * wireless Access point (APs)
- * Ethernet port in classroom & laboratories.

→ Network Topology

Hybrid Hierarchical Topology which combines :

- 1.) Tree Topology
- 2.) Extended star Topology.
- 3.) Ring Topology (server room)

→ The network hierarchical layer

- ① core layer
- ② Distribution layer
- ③ Access layer

→ core layer

* core layer contains server room and server room have firewall, router, 2-L3t switches, patch Rack, server, inverter etc.

→ Distribution layer

* Distribution layer contains 4-switches.

* The 4 switches are →

- ① Main Build → R₇ (Room no- 213, 2nd floor)
- ② Main Build → R₁ (Room no- 301, 3rd floor)
- ③ Main Build → R₆ (Room no- 113, 1st floor)
- ④ Main Build → R₂ (Room no- 201, 2nd floor)

* This 4-switches contains →

- ① 18 port Gigabit Easy smart switch with 16-port. TL-SG1218MPE
- ② SG3428XMP → 24 port Gigabit & 4-port 10GE SFP+, L2+ managed

→ Now let's discuss these 2 above components: —

① TP-Link TL-SG1218MPE

- * Likely used for: CCTV camera, wifi access point etc.
- * 18 port Gigabit "Easy Smart" switch.
- * 16 port support PoE+

② TP-Link Omada SG3428XMP

- * Higher-end managed layer 2+ switch.
- * 24 PoE+ port, uplinks, 10 G SFP
- * usually acts as a main distribution for each floor.

Access Layer :-

- * Access layer contain 3-switches i.e.

- 1.) Main Build R1 - [Room no-301, 3rd floor]
- 2.) Main Build R4 - [Room no-05, Ground floor]
- 3.) Main Build R5 - [Room no-18, Ground floor]

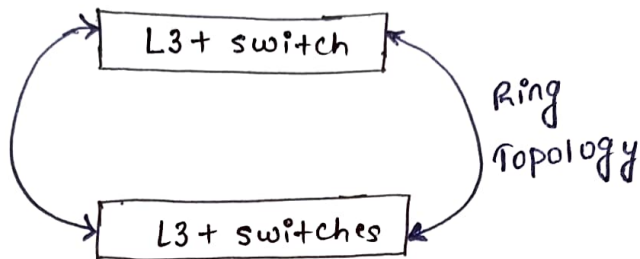
- * Every switches act as access layer which helps to provide network connectivity to all classroom, Lab, Faculty Room, etc.

- * Access layer directly connect :

- ① Ethernet port
- ② wireless Ap
- ③ Lab system
- ④ other wireless Ap (ex-OLB)

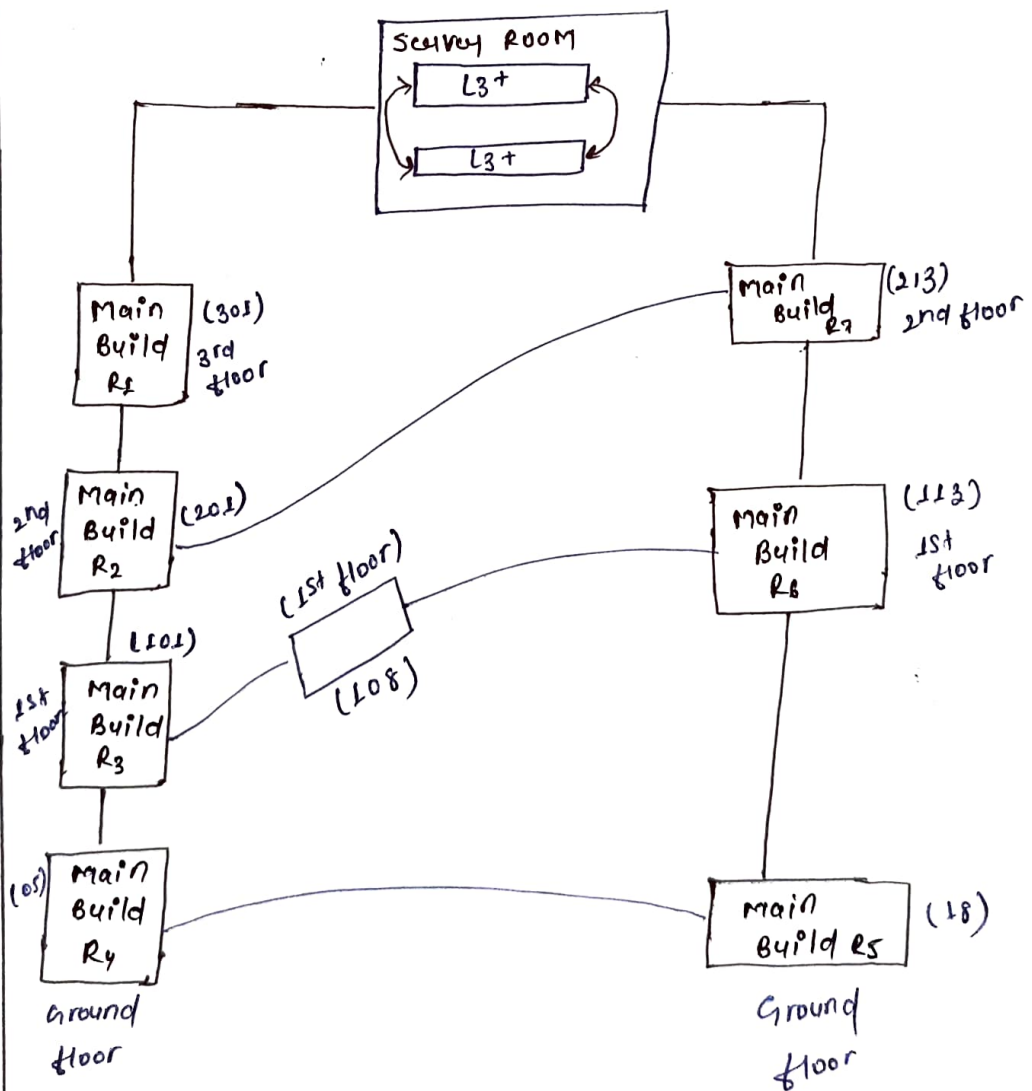
SERVER ROOM Architecture / Topology

ROUTER



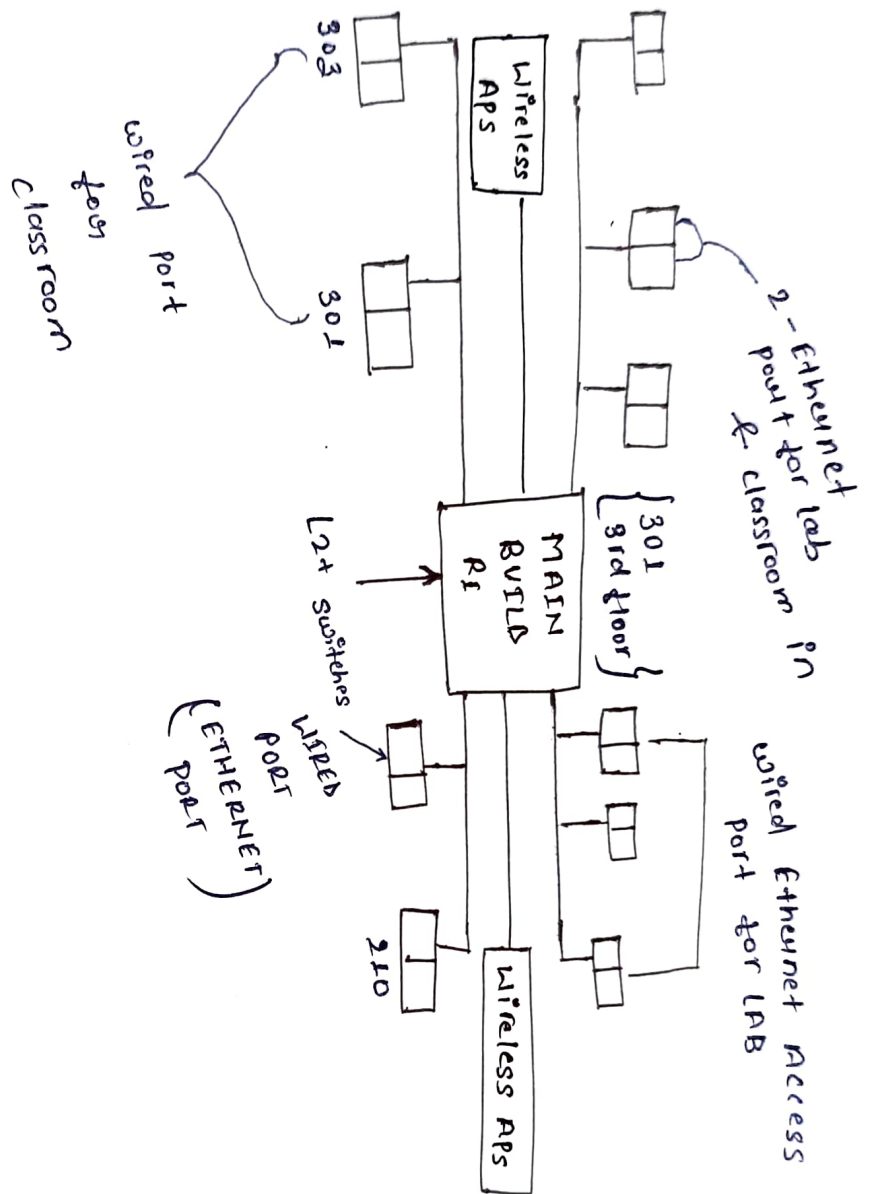
L2+ switches

Topology



- R1, R7, R2, R6 → Distribution layer
- server room → core layer
- R1, R2, R3, R4, R5, R6, R7 → Access layer

ACCESS LAYER SWITCH (3rd floor - 301)



Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\DELL> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . . :
Link-local IPv6 Address : fe80::6fca:531a:400b:3e7b%41
IPv4 Address. : 10.60.18.214
Subnet Mask : 255.255.255.0
Default Gateway : 10.60.18.98

Wireless LAN adapter Local Area Connection* 11:

Media State : Media disconnected
Connection-specific DNS Suffix . . :

Wireless LAN adapter Local Area Connection* 12:

Media State : Media disconnected
Connection-specific DNS Suffix . . :

Wireless LAN adapter Wi-Fi:

Media State : Media disconnected
Connection-specific DNS Suffix . . :

PS C:\Users\DELL> |

**CPU**

3% 1.60 GHz

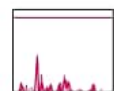
**Memory**

8.1/15.7 GB (52%)

**Disk 0 (C: E: D: F:)**

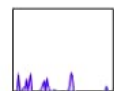
SSD (NVMe)

11%

**Ethernet**

Ethernet 2

S: 0 R: 0 Kbps

**GPU 0**

Intel(R) Iris(R) Xe Gra...

0%

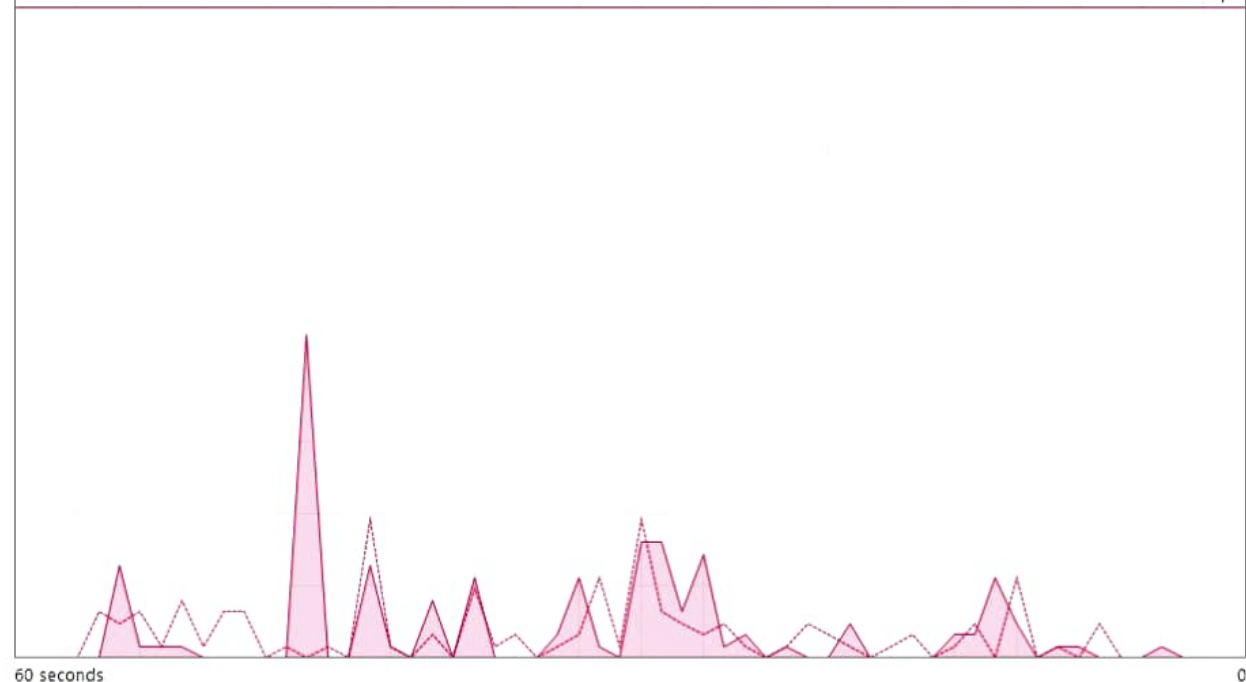
Ethernet

Remote NDIS based Internet Sharing Device

Throughput

500 Kbps

450 Kbps



60 seconds

0

Send
0 Kbps

Receive
0 Kbps

Adapter name: **Ethernet 2**Connection type: **Ethernet**IPv4 address: **10.60.18.214**IPv6 address: **2409:40e4:114b:de9b:e1c7:3588:b9d1:a121**

Windows PowerShell
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Windows IP Configuration

Wireless LAN adapter Local Area Connection* 11:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 12:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. : 2409:40e4:131b:66c7:c991:3c3:964c:aaba
Temporary IPv6 Address. : 2409:40e4:131b:66c7:95ff:830f:4fd6:4fd0
Link-local IPv6 Address : fe80::cdb6:245a:a163:49ca%10
IPv4 Address. : 10.86.89.193
Subnet Mask : 255.255.255.0
Default Gateway : fe80::54f6:b6ff:fe1a:979c%10
10.86.89.169

PS C:\Users\DELL>

Difference b/w wired & wireless speed?

Solⁿ:-

<u>Wired Network</u>	<u>Wireless Network</u>
* faster communication	* slower communication
* stable connection	* Affected by distance / Interference
* low latency	* Higher latency
* Dedicated physical medium	* shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance

→ wireless connection uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q. Role of Subnet mask in Network?

Solⁿ:- A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network

* It is important for:-

- ① Network segment
- ② Efficient routing
- ③ Communication b/w devices

OBSERVATION

During the study of the college network infrastructure : —

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed Tp-Link PoE+ switches were used in network cabinets.

Advantage of Existing Network?

- * scalable architecture
- * centralized management
- * supports PoE+ device.
- * Easy Expansion of floors & classroom
- * support both wireless & wired.
- * Redundant link for failover.

CONCLUSION

The observed campus LAN follow a Hybrid hierarchical topology consist of Tree, extended star, Ring.

The network uses managed PoE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.